Concern Resolution Process for In-house Manufacturing

QC Story Project

TO ELIMINATE THE ISSUE OF FRONT AXLE BOLT SLIP/HALF TORQUE IN SCDC 4WD VEHICLES

QC STORY PROJECT

PROJECT TITLE :-

TO ELIMINATE THE ISSUE OF FRONT AXLE BOLT SLIP/HALF TORQUE IN SCDC 4WD VEHICLES

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	01.07.18	10.07.18
Completion Date	30.10.18	25.10.18

Current Status:

closed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarize & Plan future actions
- Guidelines to user

2. Justify the choice (reasons)

PROMISE 2022
AUTOMOTIVE SECTOR

We Will Grow

Revenue 75% | Profit 100%

By

'Enhancing lifestyle' in personal segment
'Enriching livelihood' in commercial segment

Being #1 in UV, SCV, EV and #3 in HCV
Doubling volume from overseas markets

Leading in next generation mobility solutions

#InspiringPeopleToRise

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

Cross Functional Team

No.	Description	Name	Department	Mobile Number
1	Team Leader	Suryabhan Sor	Production	9763003635
2	Mentor	Taj Shaikh	Production	8378991344
3	Member 1	Satish Pagar	MQA	9822209653
4	Member 2	Mangesh Musale	MQA	9730163917
5	Member 3	Sandip Borse	Production	9753139602
6	Member 4	S A Patel / S T Pagare	Production	9960003229/ 8830974597

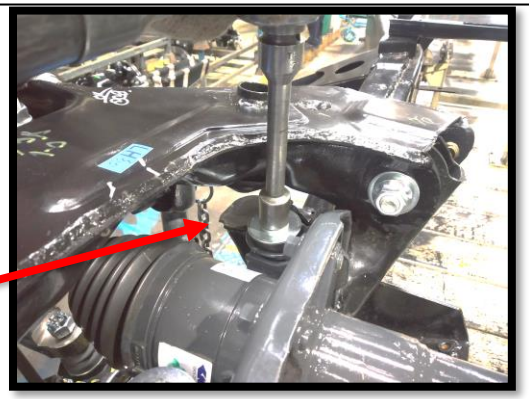
1. Select the Theme

Brief description of Problem

- What : 4WD front axle bolt slip/half torque.
- Where : At front axle drop stage.
- When : From start of W109 SCDC vehicles.
- Who : By front axle bolt torquing associate.
- Why : Repeat & major rework concern at stage and buyoff.
- How : Average 25 concerns per month.

Front axle bolt half torque.

Front axle bolt slip



2. Justify the choice (reasons)

Problem Criticality (Severity)

- **Importance & Severity of Concern :-**

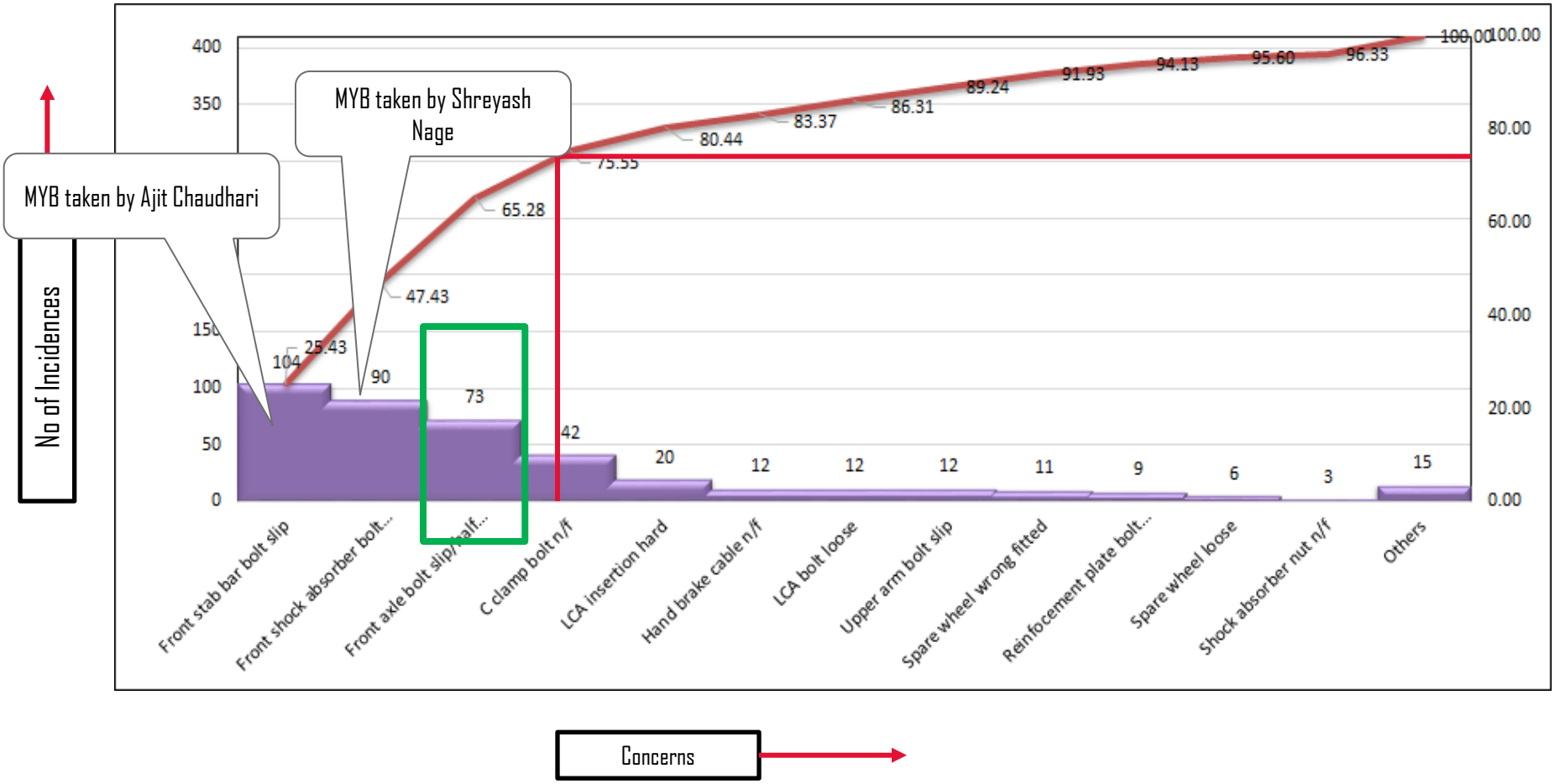
- **Front axle bolt slip is a V1 defect and axle may fall down in running condition.**
- **Major contributor in productivity & FRC loss concern.**
- **Skill operator required for rework.**
- **Rework activity is critical & time consuming.**
- **Repair cost is very high i.e. 3,05,651 per year.**

Rework Cost Calculations :-

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	120
Avg Concerns/Day	Nos	2.0
Total Required Rework time per day	Min	240
Effective working time	Min	465
Operator required /day	nos	0.52
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	1018.84
Avg no. of working days	days	25
Operator cost /month	Rs	25471.0
Operator Cost/year	Rs	305651.6

2. Justify the choice (reasons)

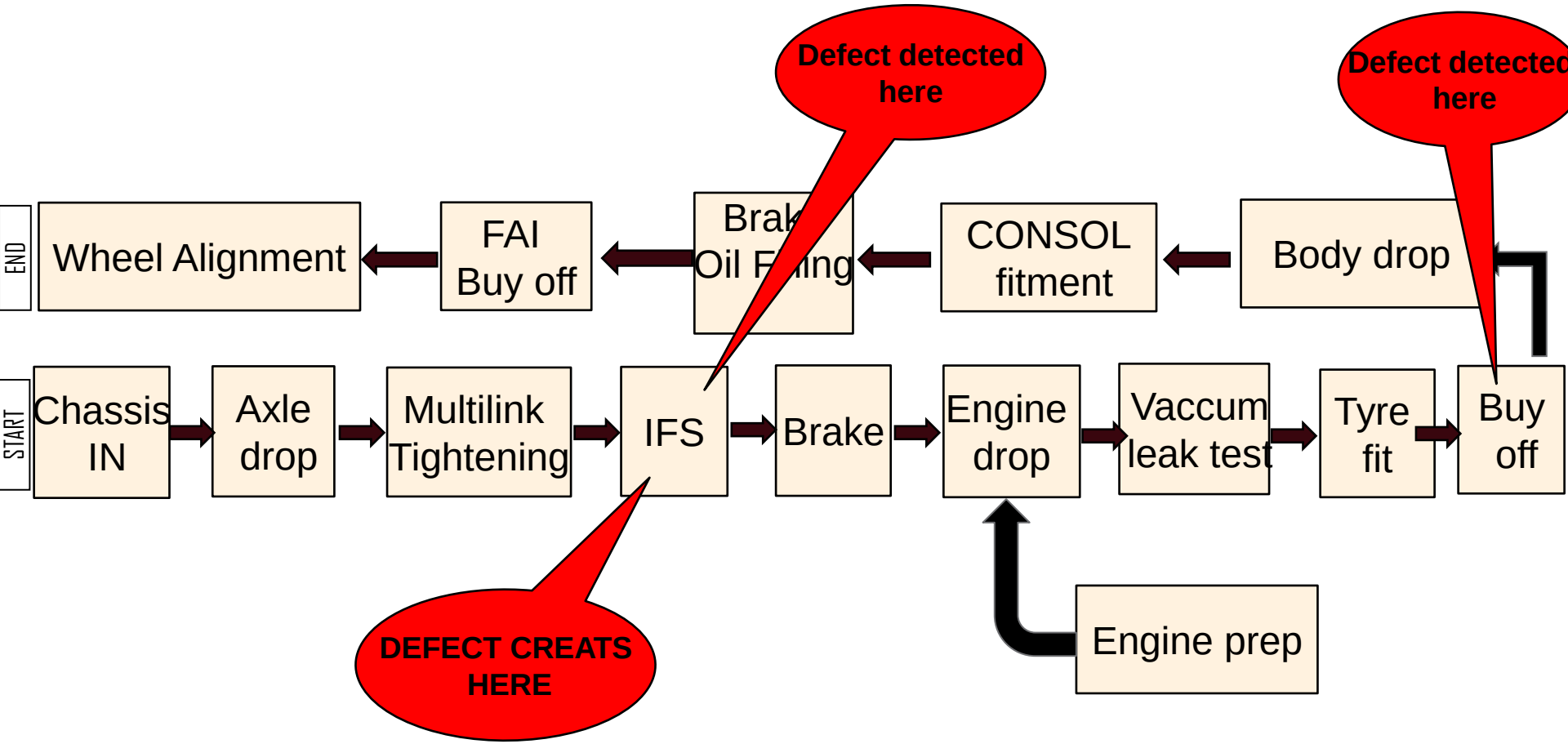
Pareto of June to Aug'19



Inference – After Pareto analysis it is observed that Front axle bolt slip concern is among top 3 concerns in QDMS Buy off

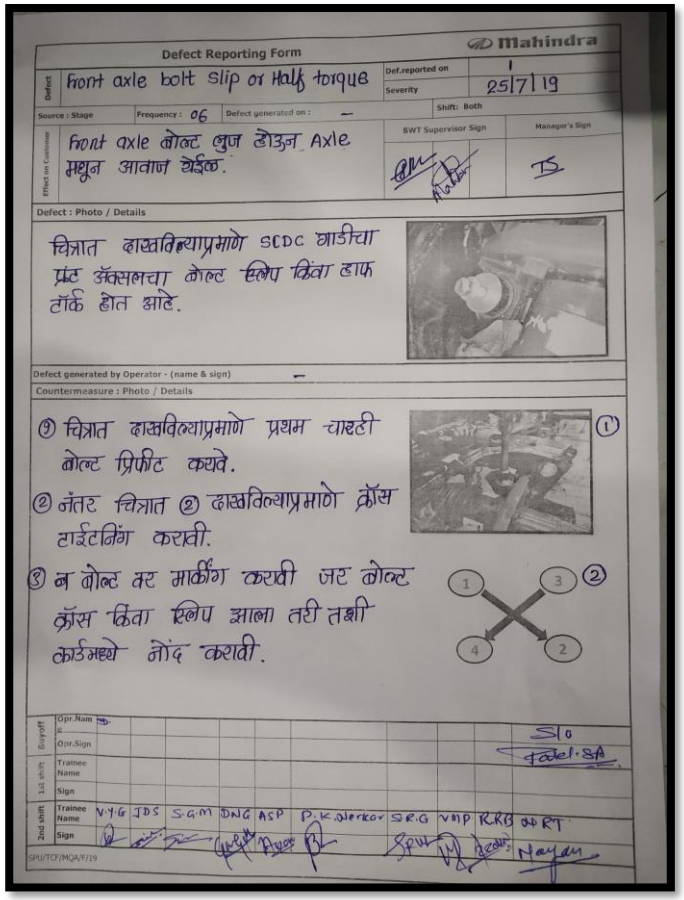
3. Understand the current situation

Observation of process



3.a Immediate Containment Action

- DRF & OPL given to operators for Concern detection at stage & buyoff.



Defect Reporting Form — Mahindra

Defect: Front axle bolt Slip or Half torque | Def.reported on: 25/7/19 | Severity: | Source: Stage | Frequency: 06 | Defect generated on: – | Shift: Both

Effect on Customer: Front axle बोल्ट तुटुन Axle मधून आवाज येईल.

BWT Supervisor Sign | Manager's Sign

Defect : Photo / Details

चित्रात दाखविल्याप्रमाणे SCDC गाडीचा फ्रंट ॲक्सलचा बोल्ट स्लिप किंवा हाफ टॉर्क होत आहे.

Defect generated by Operator - (name & sign) –

Countermeasure : Photo / Details

① चित्रात दाखविल्याप्रमाणे प्रथम चारही बोल्ट प्रिफीट करावे.

② नंतर चित्रात ② दाखविल्याप्रमाणे क्रॉस टाईटनिंग करावी.

③ न बोल्ट वर मार्किंग करावी जर बोल्ट क्रॉस किंवा स्लिप झाला तरी तशी कार्डमध्ये नोंद करावी.

Buyoff: Opr.Name, Opr.Sign — S/O Fadel SA

1st shift: Trainee Name, Sign

2nd shift: Trainee Name, Sign — V.Y.G | J.D.S | S.G.M | D.N.G | A.S.P | P.K.Sherkar | S.R.G | V.N.P | K.R.B | D.R.T

SPU/TCF/MQA/F/19

Defect communication done through DRF to all stage associate. & Buy off associate.

3. Understand the current situation

Observation of Front axle bolt tightning

Front axle lifting

Front axle drop on chassis

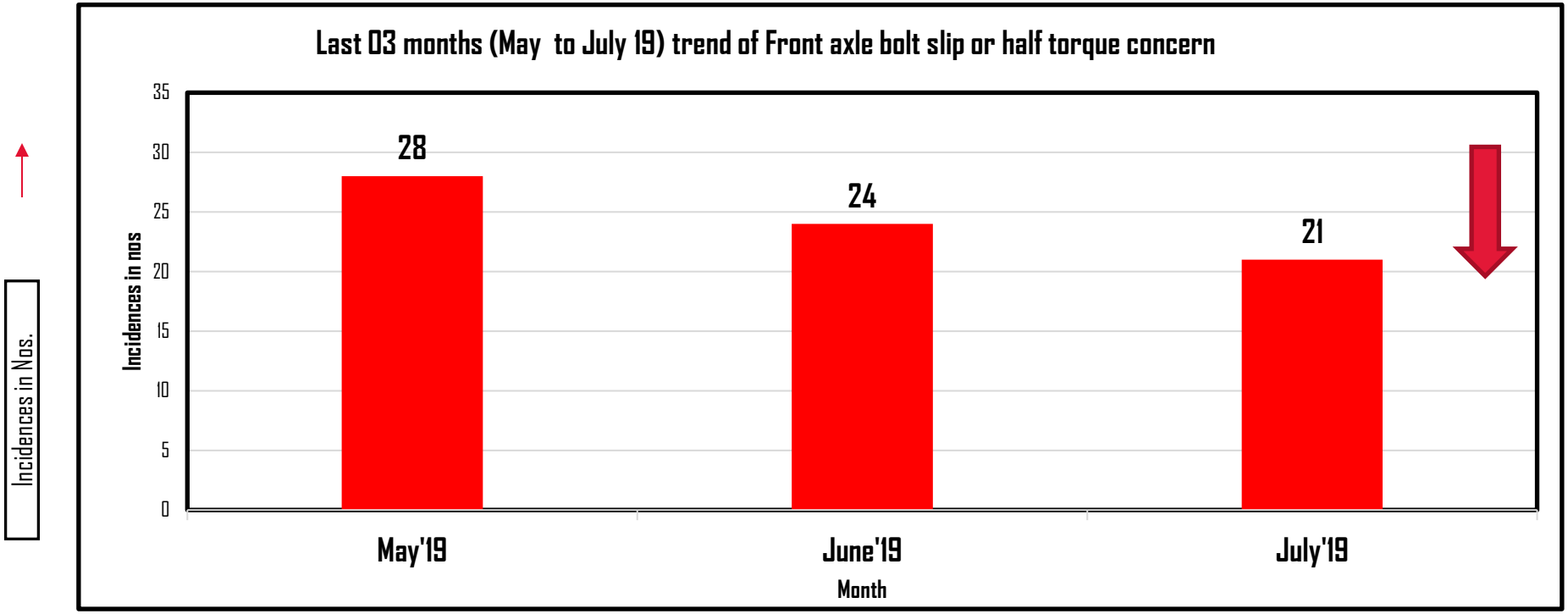
Front axle bolt prefitment

Front axle bolt torquing

Front propeller shaft fitment

Torsion bar fitment

3. Understand the current situation



DPT of Front axle bolt slip is not ok (June to Aug '19) = 495.16 R/1000.

Months

Inference – Average 25 concern per month reported at Chassis line axle drop stage

3. Understand the current situation

L0 for Vaccum leak test fail in Marazzo model.

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : Scorpio/Xylo TCF
Cell / BWT / Line: BWT 7

Requested By - MQA | Date - 14.07.2019 | Source - Vaccum test m/c | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : Front axle bolt slip or half torque

FMEA Severity /Gravity : ☐ v1+ ☑ v1 ☐ v2 ☐ v3 | No of Incidences on the day (Including Yard check) : 2-3 Vehicles/day | VIN / Aggre. No.- | Model : SCDC 4WD

Date of Incidence/ Assy : 14.07.2019 | Shift / Time : Both | Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☑ CC ☐ SC

Immediate containment action for Customer Protection :
Rectified with new parts

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - BWT . :

Does a control already exist ? ☐ YES ☑ NO

Stage / Post Analysis -

Operator Name : M S Jadhav/J D Suryawanshi

Man(Human Action)

- Is operator trained ? ☑ YES ☐ NO
- Operator level (I.L.U) : Black L
- OR Dexterity level(1...4) : L3
- Is operator under training ? ☐ YES ☑ NO
- Is operator temporary for Stage ? ☐ YES ☑ NO
- Bad rework ? ☐ YES ☑ NO

Method(process)

- Main steps are respected ? ☑ YES ☐ NO
- Is there a related key point in SOP / SOS? ☑ YES ☐ NO
- Are the reasons for key point known ? ☑ YES ☐ NO
- Are operating reqmnts respected ? ☐ YES ☑ NO
- Is the defect part of the Buy off / Checkman list ? ☑ YES ☐ NO
- Was it detected by Inspector/ Buoff Opr ? ☑ YES ☐ NO

Other

- Is there a Control Plan ? (Record in Analysis table) ☑ YES ☐ NO
- Is the packaging standard defined for the part ? ☑ YES ☐ NO
- Is the packaging standard respected ? ☑ YES ☐ NO
- Is there any alert mechanism for failure ? ☑ YES ☐ NO

Machine

- Is the specified /Tool / Fixture facility used ? ☑ YES ☐ NO
- Last calibration dates
- Facility /Tool / Fixture in good condition ? ☑ YES ☐ NO
- Is it a substitute facility ? ☐ YES ☑ NO
- Are machine parameters known ? (Record in Analysis table) ☑ YES ☐ NO
- Parameters are respected ? ☑ YES ☐ NO

Product Failure

- In only one shift ? ☐ YES ☑ NO
- In only one line ? ☑ YES ☐ NO
- The problem follows the vehicle ? ☐ YES ☑ NO
- The problem follows the part ? ☑ YES ☐ NO
- Is it a repeat problem ? ☑ YES ☐ NO
- Is the suspected parameter measurable ? (Record in Analysis table) ☐ YES ☐ NO

Material/ Consumable

- Is the specified product used ? ☐ YES ☑ NO
- Did it evolve / Change ? ☐ YES ☑ NO
- Is the storage SOP respected ? ☑ YES ☐ NO
- Is the shelf life / expiry date respected ? ☑ YES ☐ NO
- Is the part as per FIFO? ☑ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusionon Gemba (Attach Sketch/ Photograph if necessary) -
Vehicle reported for - Front axle bolt slip or half torque at IFS stage.
Vehicle reworked by - replacing front axle or by putting dummy weld nut.
History - No
Process flow -Front axle drop on chassis - all 4 bolts tightning - LCA and UCA fitment- Chassis inversion - Brake - Buyoff

Detection - Check point at chassis buyoff
Traveler card - Concern wrtten in traveller card

Packaging - Front axle supply on line & Chassis supply at In stage
Defect history - Nil

Failed part : Wwld nut spatter hown to KLT QRE as well as SQA ICA officer.

Sno	Product / Process / Machine / Parameter	Spec.	Actual
1	Angulat tool	67 NM	67 NM

Is 4M Analysis Part 2 required ? ☑ YES ☐ NO

Is the substitute transport mode used ? ☐ YES ☑ NO | **Is the packaging defined ?** ☑ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
1. DRF given to stage and buyoff associates. 2. 100% weld nut rertapping started before chassis in.	SB/AC	14.07.19	Closed

2. Permanent Action -	Resp	Date	Status
1. Part L1 required 2. Chassis weld nut spatter ICA and PCA to be revert. 3) Tightning sequence to be modify. 4) Weld nut deletion to be done same as W103 SCDC chassis.	Satish/Sor		Open

Analysed by Name/s - Suryabhan
Date - 14.07.19

Choice : ☑ WSS ☐ QC Story

- Inference from L0
Problem follows part
- Problem is a repeat problem
L1 analysis required.

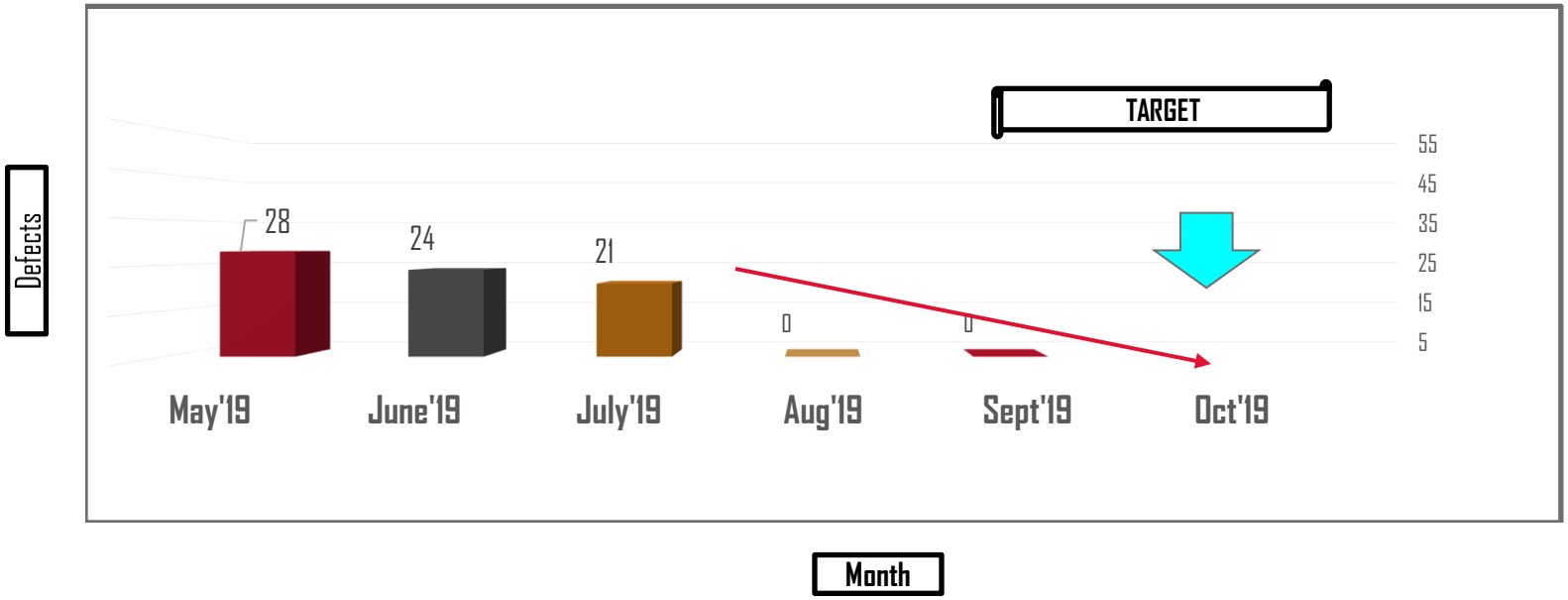
4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3/L4
Method	Front axle all four bolts tightening sequence not ok
Others	Weld nut not required same as W103 SCDC vehicles
Machine	Torque found ok
Product failure	Problem follows part
Material	Material stored as per requirement.

4. Select Target

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	Average 25 vehicles reported for front axle bolt slip or half torque in a month
Target:	Eliminate the issue of front axle bolt slip /half torque in 4WD scdc vehicles up to Oct'19.

To eliminate Front axle bolt slip/half torque concern up to Oct'19



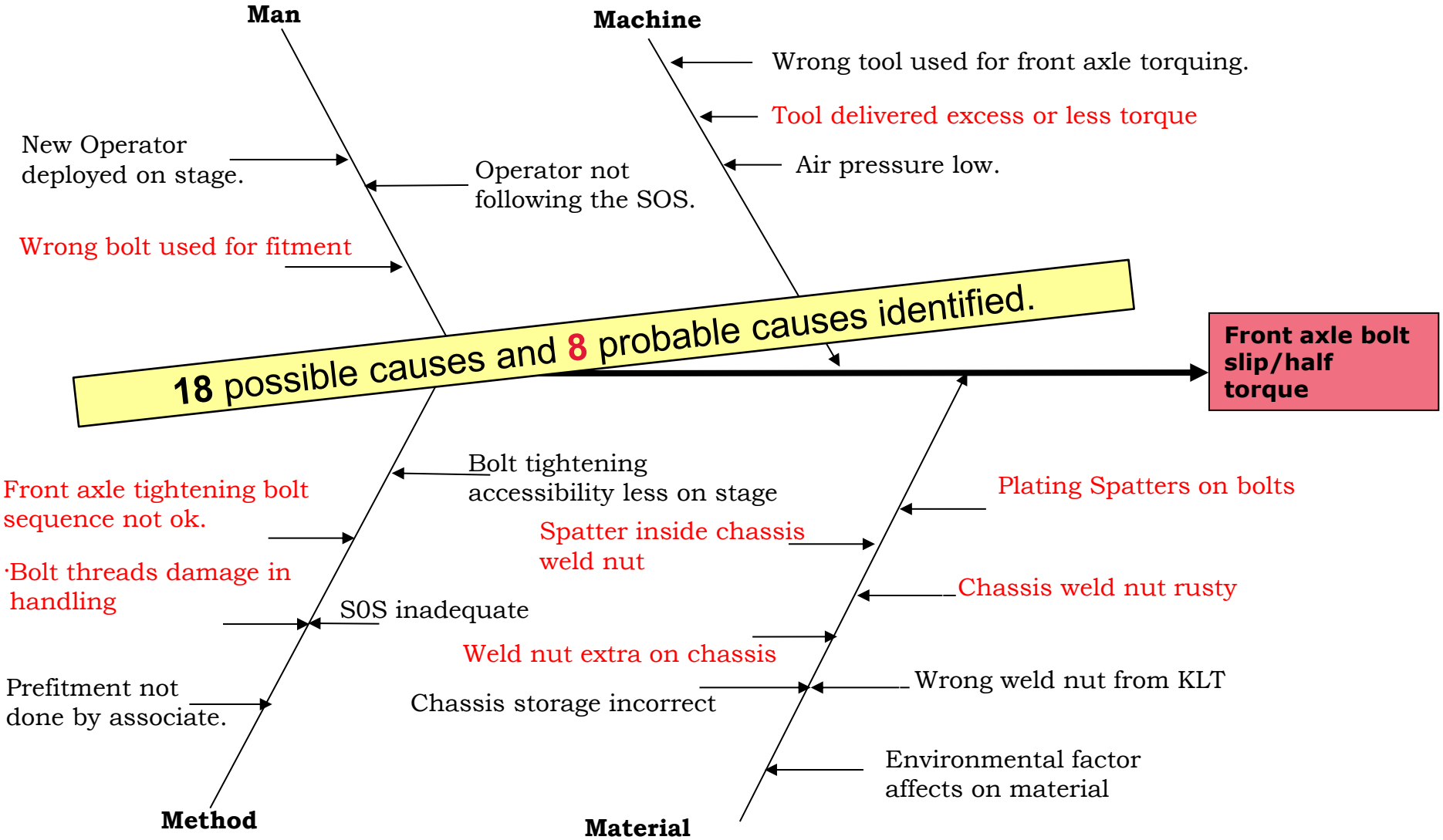
5. Analysis

List of Possible causes after brainstorming among team members

Sr. no	Possible Causes
1	New operator deployed at stage.
2	SOS Not Followed .
3	Wrong bolt used for fitment
4	Wrong tool used for front axle torquing.
5	Tool delivered excess or less torque
6	Air pressure low.
7	Front axle tightening bolt sequence not ok.
8	Bolt threads damage in handling
9	Prefitment not done by associate.
10	Bolt tightening accessibility less on stage
11	S0S inadequate
12	Spatter inside chassis weld nut
13	Weld nut extra on chassis
14	Chassis storage incorrect
15	Plating Spatters on bolts
16	Chassis weld nut rusty
17	Wrong weld nut from KLT
18	Environmental factor affect the vaccum fail.

5. Analysis

5.1 Identify possible causes: Fish Bone diagram.



5. Analysis

5.1 List of possible causes

Sr. no	Possible Causes	Suryabhan	Taj shaikh	Mangesh	Satish	Patel	Pagare	Naved	Total
1	New operator deployed at stage.	1	1	3	3	3	3	3	17
2	SOS Not Followed .	1	1	1	1	1	1	1	7
3	Wrong bolt used for fitment	3	3	9	3	3	9	9	39
4	Wrong tool used for front axle torquing.	3	1	3	1	3	1	1	13
5	Tool delivered excess or less torque	3	9	3	3	3	9	9	39
6	Air pressure low.	1	3	3	3	3	3	3	19
7	Front axle tightening bolt sequence not ok.	9	9	3	9	9	3	3	45
8	Bolt threads damage in handling	3	3	9	9	9	3	9	45
9	Prefitment not done by associate.	1	1	3	1	3	1	1	11
10	Bolt tightening accessibility less on stage	1	3	1	1	3	3	3	15
11	SOS inadequate	3	3	3	1	3	1	1	15
12	Spatter inside chassis weld nut	3	9	3	9	3	9	9	45
13	Weld nut extra on chassis	9	9	9	3	9	9	3	51
14	Chassis storage incorrect	1	1	3	3	3	3	3	17
15	Plating Spatters on bolts	9	3	6	3	6	9	6	42
16	Chassis weld nut rusty	3	9	9	9	3	9	3	45
17	Wrong weld nut from KLT	1	1	3	1	3	1	1	11
18	Environmental factor affect the vaccum fail.	1	3	1	1	3	3	3	15

Scale: 0=None, 1=Low, 3=Moderate, 9=Strong
Threshold Limit = 39

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr.No	Not considered Causes	Reasons
1	New operator deployed at stage.	All trained operator deployed on stage.
2	SOS Not Followed .	All SOS follows the SOS of bolt tightning
3	Wrong tool used for front axle torquing.	Correct tool with 67Nm used.
4	Air pressure low.	Air pressure found 6 bar
5	Prefitment not done by associate.	Visually checked confirmed OK.
6	Bolt tightening accessibility less on stage	Accessibility Checked & found ok
7	Chassis storage incorrect	Chassis stored in good condition
8	Wrong weld nut from KLT	Weld nut found as per spec.
9	Environmental factor affect the vaccum fail.	Checked & found ok
10	SOS inadequte	All points found ok

5.1 TEST OF HYPOTHESIS

Confirm the probable cause:-

Sr.No	Probable Cause
1	Wrong bolt used for fitment
2	Tool delivered excess or less torque
3	Front axle tightening bolt sequence not ok.
4	Bolt threads damage in handling
5	Spatter inside chassis weld nut
6	Weld nut extra on chassis
7	Plating Spatters on bolts
8	Chassis weld nut rusty

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
1	Wrong bolt used for fitment.	Bolt specifications M12x1.25x80 F/B	100 No bolts checked on stage and inside KLT and all found ok.	**Hypothesis invalid**

Bolt specification on box and inside KLT

M12X1.25X80 F/B

Inference- 100 No's bolts checked physically on stage and KLT and all found ok

5.3 Confirm the probable cause:

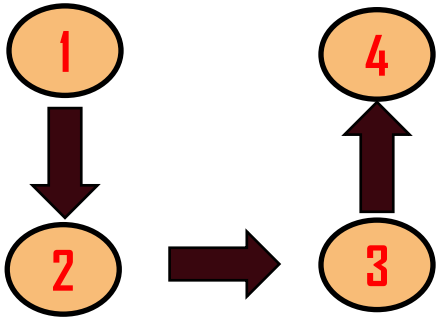
No	Probable Cause	Testing method	Observations	Conclusion
2	Tool delivered excess or less torque	Tool dynamic torque checked on ACTA	All torque value found within specification 63+/- 5 Nm	**Hypothesis invalid**

Tool dynamic torque checked on ACTA and found ok

Inference- Tool dynamic torque 10 cycles checked and found within specifications 63+/- 5 Nm

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
3	Front axle tightening bolt sequence not ok.	Visually by process audit	20 vehicles observed with sequence as per sos and out of 20 in 6 vehicles bolt half torque/slip	**Hypothesis Valid**



Tightening sequence as per SOS

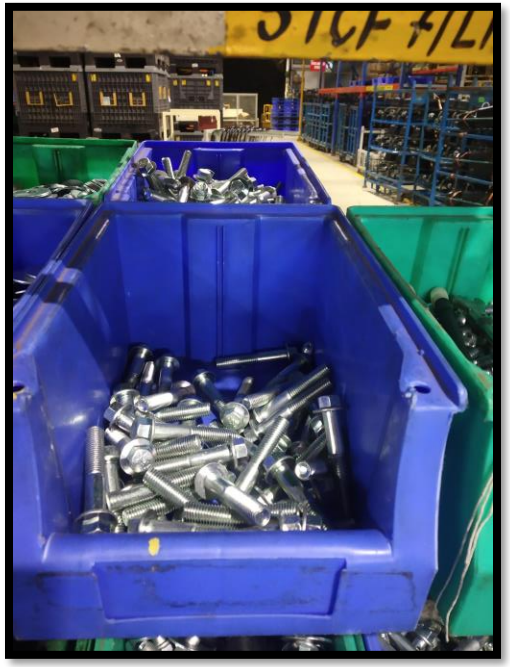
Tightening process observation with CFT

Inference – After following SOS still bolts are getting slip/half torque

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
4	Bolt threads damage in handling	Bolts thread checking by nut of same pith outside.	2 bins checked on stage as well as in KLT all bolts threading found ok	**Hypothesis Invalid**

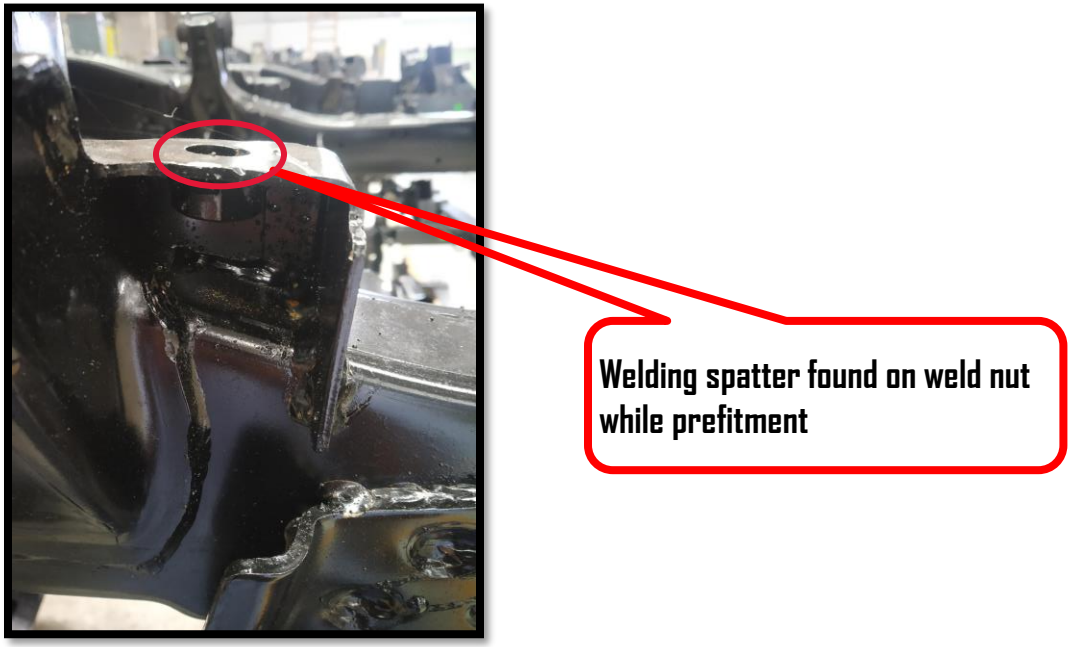
All bolts threading found ok as nut easily getting prefit on bolt threads.

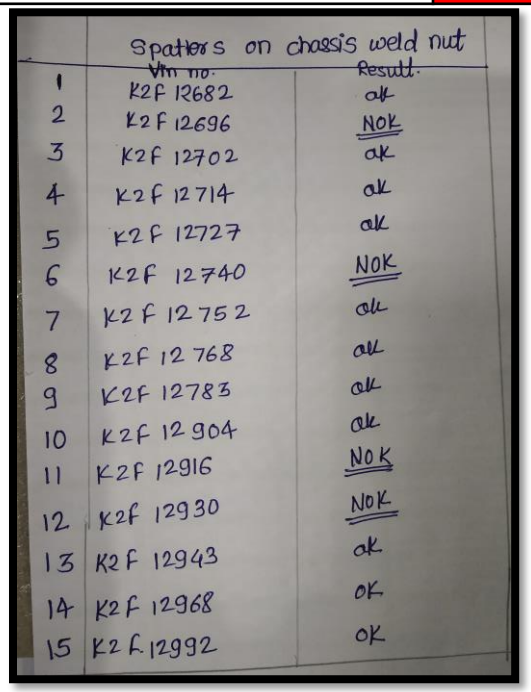


Inference- Torque wrench calibration done and found OK, hence this is a invalid cause.

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
5	Spatter inside chassis weld nut	Visual checking and by bolt pre tightening on weld nut	15 chassis weld nut checked offline and in 4 chassis spatter found on weld nut	**Hypothesis Valid**





Spatters on chassis weld nut

	Vin no.	Result.
1	K2F 12682	ok
2	K2F 12696	NOK
3	K2F 12702	ok
4	K2F 12714	ok
5	K2F 12727	ok
6	K2F 12740	NOK
7	K2F 12752	ok
8	K2F 12768	ok
9	K2F 12783	ok
10	K2F 12904	ok
11	K2F 12916	NOK
12	K2F 12930	NOK
13	K2F 12943	ok
14	K2F 12968	OK
15	K2F 12992	OK

Inference- 15 chassis checked and in 4 chassis welding spatter found on weld nut

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
6	Chassis weld nut design inadequate	Comparison of W109,W109 and W110, Thar and Bolero 4WD chassis.	In every W109 chassis there is weld nut in 100% chassis where as there was no weld nut on W103 chassis.	**Hypothesis Valid**

Benchmarking -
With benchmarking of W103 and Thar 4WD chassis there is no weld nut for front axle bolts and hence following are the benefits –

1. Bolt mounting profile is straight.
2. Weld nut offset concern eliminated
3. No spatter on weld nut
4. No slippage or half torque of bolt.

Inference – Excluding W109 SCDC chassis there wad no weld nut for any 4WD chassis front axle bolts.

5.3 Confirm the probable cause:

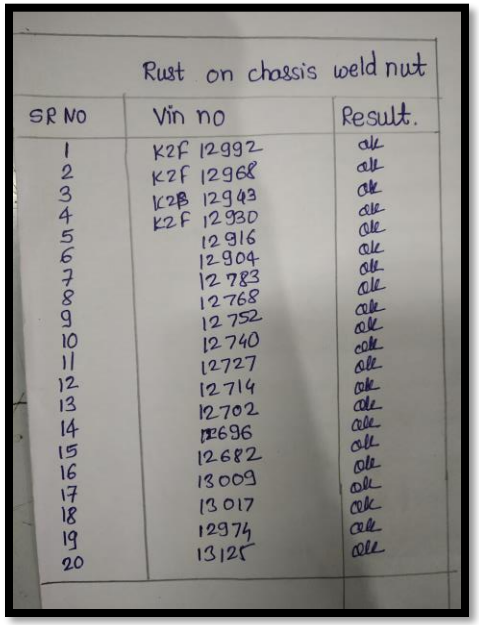
No	Probable Cause	Testing method	Observations	Conclusion
7	Plating Spatters on bolts	Process audit by visual check	50 no's bolts checked visually for plating spatters but all bolts found ok	**Hypothesis Invalid**

All bolts found ok for no plating spatter on bolts

Inference- Out of 50 bolts all bolt found ok for plating spatter.

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
8	Chassis weld nut rusty	Visually and by bolt pre fitment	Out of 20 chassis in no chassis weld nut found rusty and bolt pre fitment was found ok	**Hypothesis invalid**



Rust on chassis weld nut

SR NO	Vin no	Result.
1	K2F 12992	ok
2	K2F 12968	ok
3	K2B 12943	ok
4	K2F 12930	ok
5	12916	ok
6	12904	ok
7	12783	ok
8	12768	ok
9	12752	ok
10	12740	ok
11	12727	ok
12	12714	ok
13	12702	ok
14	12696	ok
15	12682	ok
16	13009	ok
17	13017	ok
18	12974	ok
19	13125	ok
20		

No rust on chassis weld nut and on chassis

Inference- Out of 20 chassis no chassis found with rusty weld and bolt prefitment and tightening found ok.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Wrong bolt used for fitment	Bolt specifications M12x1.25x80 F/B	**Hypothesis invalid**
2	Tool delivered excess or less torque	Torqueing done on line vehicles with CFT on stage.	**Hypothesis invalid**
3	Front axle bolt tightening sequence not ok.	Visually by process audit	**Hypothesis valid**
4	Bolt threads damage in handling	By prefitment of bolts	Hypothesis Invalid
5	Spatter inside chassis weld nut	Visual check and by bolt prefitment.	**Hypothesis valid**
6	Chassis weld nut design inadequate	By benchmarking with W103 chassis	**Hypothesis valid**
7	Plating Spatters on bolts	Process audit by visually	Hypothesis Invalid
8	Chassis weld nut rusty	By visual checking of chasis	Hypothesis Invalid

5.3 Identify the root causes: (Use Why – Why)

WHY - WHY Analysis

Sr No	Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION
1	Front axle bolt tightening sequence not ok	If one end prefit another end not getting engaged easily.	RTB mounting brackets are fixed	Brake tube profile is fixed	Current tightening sequence incorrect	W109 4WD front axle all bolts tightening sequence to be change.
2	Spatter inside chassis weld nut	Spatters coming from supplier	Skipped from inspection	Checking frequency low		Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random)
3	Chassis weld nut deign inadequate	Both side tightening direction is different.	Tightening access is in inner side of chassis and tube get came out from clip.	Brake tube profile get damaged.	Excess stress developed on caliper hose end if another end of tube tighten first.	Bench marking with W103 KLT scdc chassis in W109 scdc chassis also weld nut to be delete

6. Implement the corrective measures

Possible Remedies Identified

Legends

High ↑ Low ↓

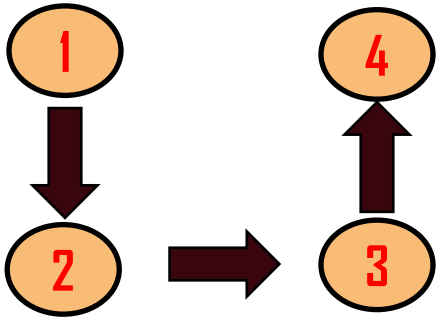
SN	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Front axle bolt tightening sequence not ok	W109 4WD front axle all bolts tightening sequence to be change.	Low	Low	Low	Medium	Accepted
2	Spatter inside chassis weld nut	Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random)	Low	Low	Low	Medium	Accepted
3	Chassis weld nut deign inadequate	Bench marking with W103 KLT scdc chassis in W109 scdc chassis also weld nut to be delete	Low	Medium	High	Low	Accepted

6. Implement the corrective measures

6.1 Actions Decided

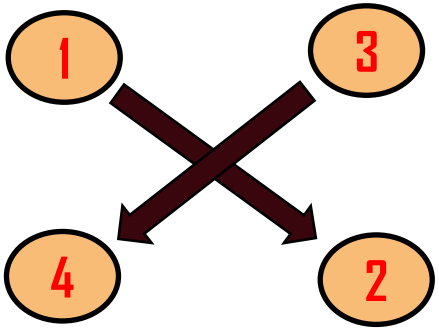
Root Cause	Action Plan	Target Dt	Resp.	Status
Front axle bolt tightening sequence not ok	W109 4WD front axle all bolts tightening sequence changed,	20 Sep 19	Mandar Padwal	Closed

Before



Tightening sequence as per SOS

After

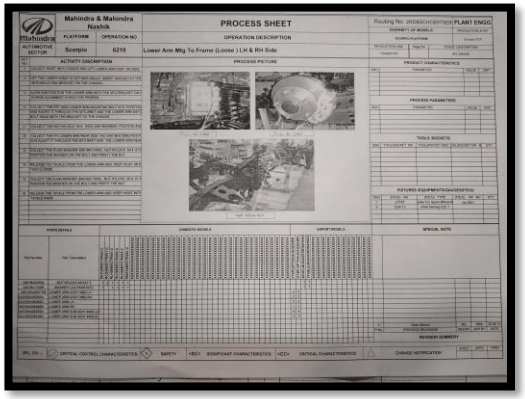


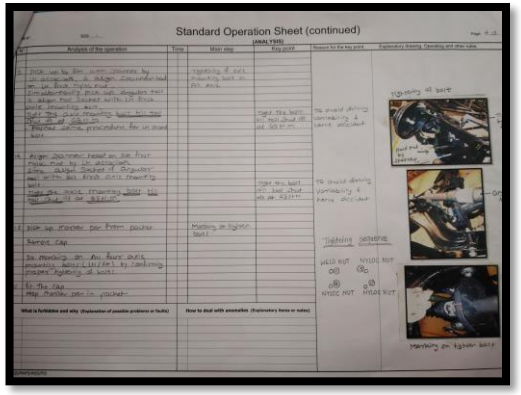
Modified Tightening sequence as per SOS

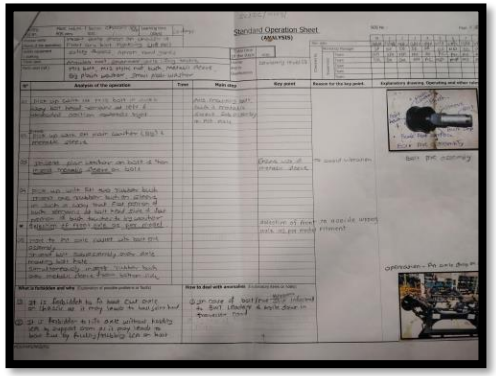
6. Implement the corrective measures

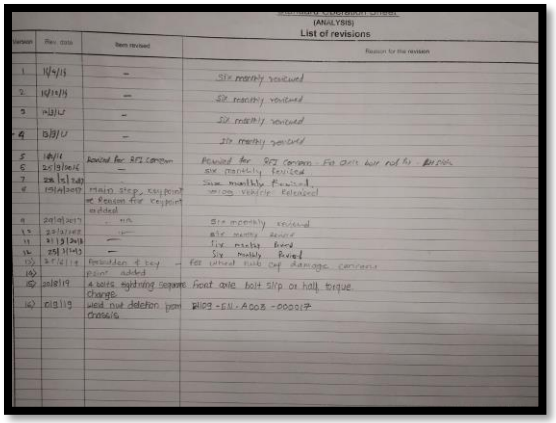
6.1 Actions Decided

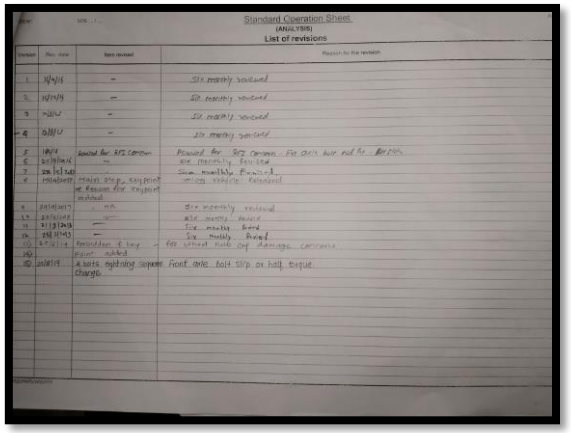
SOS











6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Spatter inside chassis weld nut	Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random)	24 Aug 19	Shailesh Kumar	Closed.

Before – Spatters on weld nut

After – No spatters on weld nuts

6. Implement the corrective measures

6.4 Actions Decided

SCAP -

F	CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)				
Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	In defective chassis bolt slip due to burr or spatter inside thread	1.Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random) 2.Corrective action taken from tier II supplier	Swapnil S.	24.08.2019	24.08.2019
2	Nut tapping not checked / done at tapping stage	Training and awareness given to operator	Amol S.	24.08.2019	24.08.2019

G ACTION TAKEN TO PREVENT RECURRENCE

1.Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random)

2.Corrective action taken from tier II supplier

3.Training and awareness given to operator

Page 2

H CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT

I Following quality documents changed due to implemented action plan :

(mention doc. no.)

Process Flow Chart	☐	Gauge Drawing	☐
PFMEA	☐	Fixture Drawing	☐
Control Plan			
Work instruction / SOP	☑		

L EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)

Are actions taken are sufficient to fix the problem	☐ YES	☐ NO
Are actions planed completed / implemented	☐ YES	☐ NO
Is problem repeated in lots received after SCAP issue Date	☐ YES	☐ NO

FOR MAHINDRA & MAHINDRA (A.S.)	SUPPLIER'S REPRESENTATIVE Yogeshwar vengurlekar
Signature / Date	Signature / Date

MQS/SCAP/ F/01 Rev.02

TO BE FILLED BY M&M

M & M Ltd. Automotive Sector	SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP)	PU Name : Scorpio Sr.No: Date: 24.08.2019

Supplier:- KLT AUTOMOBILE & TUBULAR PRODUCTS LTD — Supplier Code: DK055
Part Description: Scorpio SC/DC W109 — Part No.:0201AAM00850N
When Observed (Date / Period) : 22.08.2019 — GR/Invoice No.:

A PROBLEM STATEMEN Should Answer — ISSUE REPORTED : REPEAT ☐ / 1'ST TIME ☑

What (is the problem? (Expression by customer)	Problem Statement:
Who is affected	Front axle mounting bolt slip
Where Location	M&M Nashik
When Since	22.08.2019
Why is the problem	01
How much Qty	Nut offset

(Problem statement should answer 5W 1H ie What, When,Who,Where, & How.
(Attach sketch / data for details where required)

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)

5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
Front axle mounting bolt slip	Burr and spatter inside nut threading	Nut threading not checked at incoming inspection	Masking of bracket not done	

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:

1) Defective chassis reworked and verified at customer end

2) All stock chassis checked & verified.(In plant)

Page 1

E ROOT CAUSE ANALYSIS OF PROBLEM: ☑ YES ☐ NO

Machine
- Is the specified M/C used ? ☑ YES ☐ NO
- Is M/C in good condition ? ☑ YES ☐ NO
- Is it a substitute M/C ? ☐ YES ☑ NO
- Are M/C parameters known ? ☑ YES ☐ NO
- Parameters are respected ? ☑ YES ☐ NO

Man
- Is operator trained ? ☑ YES ☐ NO
- Is operator is under training ? ☐ YES ☑ NO
- Is operator temporary ? ☐ YES ☑ NO
- Is it due to Bad Rework ? ☐ YES ☑ NO

Measurement
- Are instruments / gauges calibrated ? ☑ YES ☐ NO
- Are Control Methods Capable to detect Non conformity? ☑ YES ☐ NO
- Are specified Control Methods used ☑ YES ☐ NO

Method
- Are Control Plan/SOP available near M/C? ☑ YES ☐ NO
- Are Control Plan respected ? ☑ YES ☐ NO
- Is the defect part was detected by operator / inspector & later mixed with production ? ☐ YES ☑ NO

Material / Tooling
- Is the specified Raw Matl./Tooling used ? ☑ YES ☐ NO
- Is the Raw Matl./Tooling Supplier change ? ☐ YES ☑ NO
- Is the heat treatment method change ? ☐ YES ☐ NO

Environment
- Is process needs environmental control? ☐ YES ☑ NO
- Is it implemented ? ☐ YES ☑ NO
- Does it conform ? ☐ YES ☑ NO

Analysis conclusion:-

We suspected that in defective chassis bolt slip due to burr or spatter inside thread

6. Implement the corrective measures

6.3 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Chassis weld nut deign inadequate	Bench marking done with W103 KLT scdc chassis in W109 scdc chassis also weld nut deleted	15.09.19	Satish Pagar	Closed.

BEFORE

Front axle mounting on weld nut

AFTER

No weld nut for front axle bolt mounting

AFTER

6. Implement the corrective measures

6.3 Actions Decided

EN:-

Mahindra Rise.

Engineering Notice Release: W109-EN-A003-000017
AD Product Development
Mahindra & Mahindra LTD(A.S)

Engineering Notice No : W109-EN-A003-000017 — Released Date : 23-May-

Reference Change Req No : DCR036452

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : RELEASE OF W109 SCORPIO SCDC FRAME ASSY DRAWING.

Reason : NEW MODEL PRODUCTION — Quantity : 500-(Trial anc

Reason Code : New Model / Variant/Feature release — Priority : Mandatory

Note : RELEASE OF W109 SCORPIO SCDC FRAME ASSY DRAWING.

Affected Models : W109 SCORPIO SC/DC,

Is EN Re Released? : — Parent EN NO :

Life Cycle State : Released — Owner Name : Jadhav P

Engineer : Jadhav Prashant (23072302)

Approved By : 301148-AHIRE MANOJ

Authorized By : VE-SCORPIO-HEAD_207094-GAUBA MANOJ

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

FTR -

FTR/R&D/F20/321

FITMENT TRIAL REPORT

Trial Report Serial No:- — Date:- 12/08/2019

Comments On Fitment

Please ensure that part inspection report(s) (with actual dimns) is attached.

No issues for fitment.

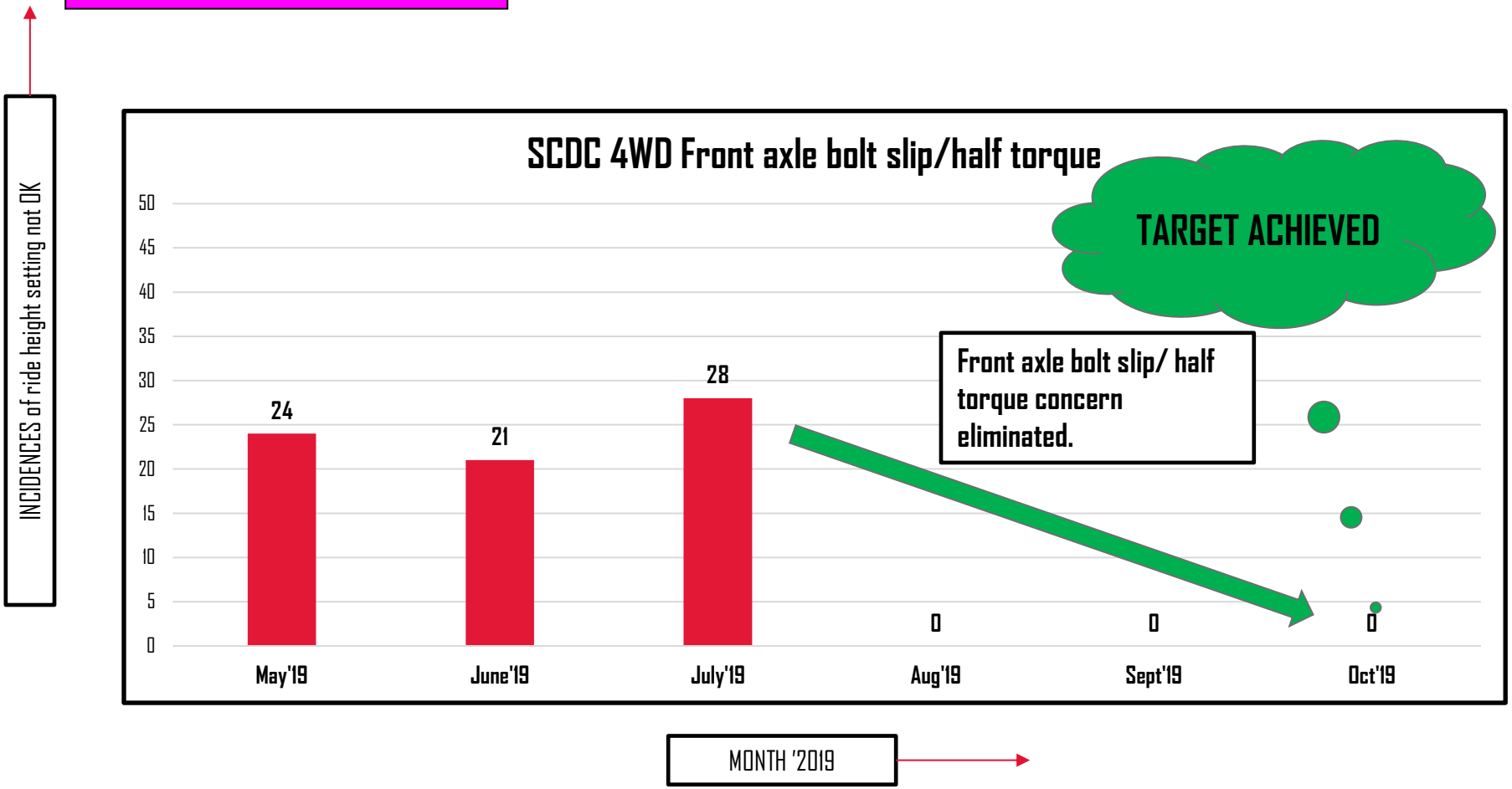
Stackup analysis to be done.

No issues for fitment. Residual torque checked found ok. PTR required.

Format No.: MQS/FPAP/Mg/F/02 — Rev: 05 — Date 30 Jan 2019

7. Check & confirm the results & effects

7.1 Verify Results:



Inference- After tacking PCA action, Front axle bolt slip/ half torque concern eliminated.

7. Check & confirm the results & effects

7.3 Full Implementation Plan :

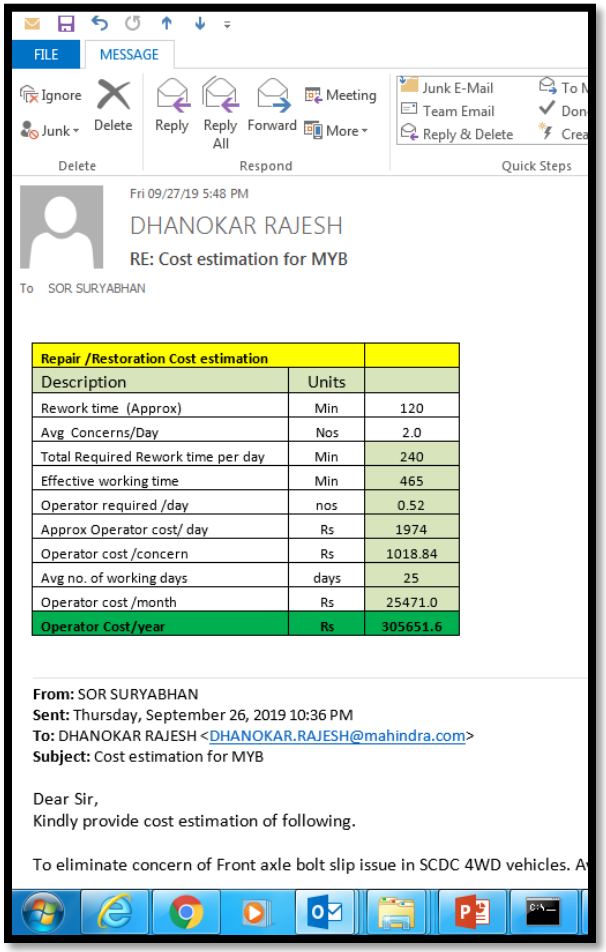
Action Plan	Target Dt	Resp.	Status
W109 4WD front axle all bolts tightening sequence changed,	20 Sep'19	Mandar Padwal	Closed
Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random)	24 Aug'19	Shailesh Kumar	Closed
Bench marking done with W103 KLT scdc chassis in W109 scdc chassis also weld nut deleted	5 Sept 19	Satish Pagar	Closed

Check & confirm the results & effects (Cont.....)

7.3 Tangible benefits:

- REWORK cost saving – Rs. 3,05,651/Year.
- Concern elimination from 20-25 cases/month to 0.

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	120
Avg Concerns/Day	Nos	2.0
Total Required Rework time per day	Min	240
Effective working time	Min	465
Operator required /day	nos	0.52
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	1018.84
Avg no. of working days	days	25
Operator cost /month	Rs	25471.0
Operator Cost/year	**Rs**	**305651.6**



7.3 Intangible benefits:

1. Operator fatigue reduced due to major rework of front axle change.
2. Final line vehicle through back eliminated.
3. Chassis line FRC increased by 5 %.

8. Standardization

8.1 Updation of documents:

FMEA- Front axle fitment on chassis

Plant : A003	Model :Scorpio	**Mahindra & Mahindra Ltd.**	Cross Functional Team
Part No/s.:	Optn No.: 5690		Nikhil / Suryabhan
Latest Change Letter:NA	Optn Desptn: Front Axle & Lower Arm Fitment (4WD only)	**Process Potenial Failure Mode & Effect Analysis.**	Vikrant ghule
PU/ Shop :STCF	M/C Name: NA	Reviewed by : Nikhil Hatedkar — Rev 23 — Doc No.: PFMEA/SPU/SCORPIO/56	Vikrant Morade
Module/ Cell : CHASSIS LINE		Approved by : Taj Shaikh — Rev 10.09.19 — FMEA Date (Orig) : 24.08.09	Hari Nagar

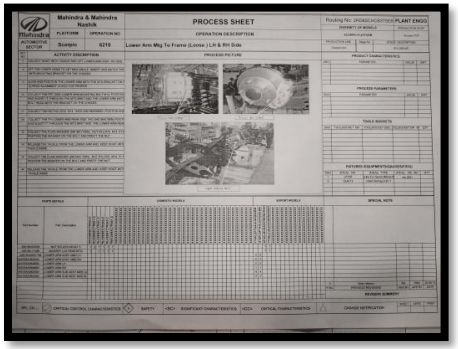
PROCESS STEP / FUNCTION	REQUIREMENTS	POTENTIAL FAILURE MODE	POTENTIAL EFFECT(S) OF FAILURE	SEV	CLASS	POTENTIAL CAUSE(S) MECHANISM(S) OF FAILURE	CURRENT PROCESS CONTROL PREVENTION	OCCUR	CURRENT PROCESS CONTROL DETECTION	DETEC	RPN	RECOMMENDED ACTION(S)	RESPONSIBILITY & TARGET COMPLETION DATE	ACTION RESULTS: ACTION TAKEN	SEV	OCC	DET	R.P.N.
Front Axle & Lower Arm Fitment (4WD only)	Frt.axle proper positioning & mounting on chassis Mtg of LCA on chassis (63+/-5 Nm)	Mounting bolt under torqued/loose.	Customer effect: 1) Squeak & rattle (9) 2) Unsafe to vehicle operation(9)	9	cc	Associate not prefit the bolt or cross tight	Only trained associate deployed on stage for bolt fitment.	3	Checkpoint at chassis buyoff.	7	189	1)Front axle bolt tightning sequence changed. 2)Front axle weld nut deletion	SS/AC Satish	Closed Closed	9	2	4	72
				9	cc	Improper power tool calibration	Calibration as per frequency decided in control plan.	2	Sticker for due date for calibration on tool.	7	126							
				9	cc	Stage Associate forget to tighten the bolt.	Only trained Associate deployed on stage.	2	Checkpoint at CAI	8	144							
		Mounting bolt missing	Customer effect: 1) Squeak & rattle (5) 2) Unsafe to vehicle operation(9)	9	cc	Stage Associate forget to tighten the bolt or fit the bolt.	Only trained Associate deployed on stage. Self check and marking on bolt head.	2	Checkpoint at CAI	8	144							
		Mounting bolt over torqued.	Assy. effect: Damage of washer	9	cc	Use of wrong power tool.	Provision of dedicated power tool set at required specification.	2	Identification sticker is provided on the tool	7	126							
				9		Improper power tool calibration	Calibration as per frequency decided in control plan.	2	Sticker for due date for calibration on tool.	7	126							
		Bolt loose	Assy. effect: 1) Loose fitment of front axle,(5) Customer effect: 2) Squeak & rattle (5) 3) Unsafe to vehicle operation(9)	9	cc	Lock nut is not getting hold by cage on chassis (Tolerence difference.	Tolerence is maintained at mean	3	At the same stage	7	189							
	Lower arm: Take the front side bolt and insert it through mtg.bracket , match the bolt head with bracket on chassis.push the bolt and prefit the nyloc nut. Position the gauge & lock it with pin & nut	Bolt head not matched with the bracket on chassis.	Assy.effect: 1) Lower arm will remain loose (5) Customer effect: 2) In running noise will come.(5) 3) Unsafe to vehicle operation(9)	9	cc	Associate does not match the bolt head with bracket	Associate training.	2	Detection at the same stage.	7	126							
	Collect front axle from pallet and check for wheel hub & then fit on to chassis.	Wrong selection of axle.Wrong gear ratio.	Customer effect: Vehicle inoperative in 4WD mode.Engagement Not possible.	8		Associate selects wrong axle having Wrong gear ratio.	Associate training & Awareness. Modelwise Matrix having Gear Ratio display on Stage.	2	No current process control detection	5	80							0
		Wheel hub damage	Customer effect: Rework	4		Associate select axle with damage wheel hub	Associate training & Awareness.	4	Self check	7	112							

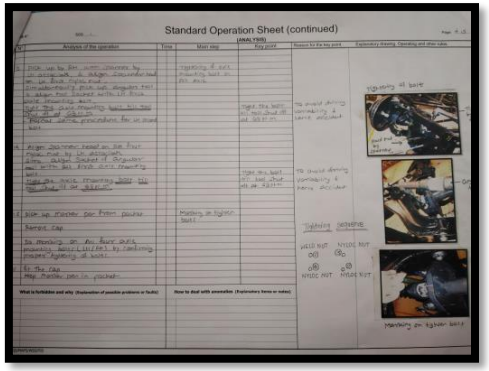
Rev No.:	Rev Date	Revision Summary	Reason for Revision
0	13.02.10	Revision after VP0 & ongoing VP built	Revision after VP0 & ongoing VP built
1	13.05.10	Revision end of VP	Revision end of VP
2	08.12.10	End of PP01 built and ongoing PP02 built	End of PP01 built and ongoing PP02 built
3	04.02.11	No change	Document reviewed
4	30.10.11	no change	Document reviewed
5	24.03.12	no change	Document reviewed
6	30.03.12	Wrong gear ration axle fittment on vehicle.	Document reviewed for SCR in March12
7	01.04.12	Wrong gear ration axle fittment on vehicle.	Document reviewed for SCR in April'12
8	20.09.12	no change	Document reviewed
9	29.04.13	No change	Six monthly review
10	30.12.13	No change	Six monthly review
11	17.10.14	No change	Six monthly review
12	30.03.15	No change	Revised for Scorpio SCDC
13	30.11.15	No change	Six monthly review
14	18.02.16	Retorquing Strated	Export RFI V1 Concern - axle mounting bolt missing.
15	26.10.16	No change	Six monthly review
16	25.04.17	No change	Six monthly review
17	13.10.17	No change	Six monthly review
18	20.2.18	No change	Six monthly review
19	7.8.18	No change	Six monthly review
20	12.2.19	No change	Six monthly review
21	5.8.19	No change	Six monthly review
22	20.08.19	Front axle bolts tightning sequence changed	Font axle bolt slip or half torque issue
23	10.09.19	Front axle mounting weld nut deletion	W109-EN-003-00017_001

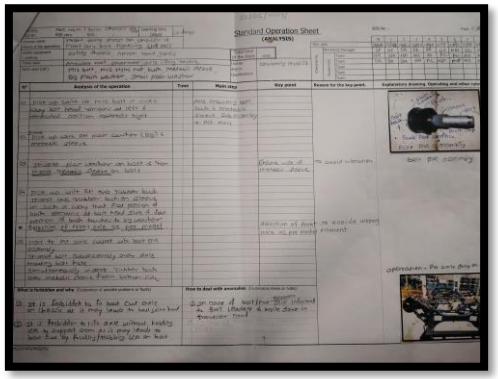
8. Standardization

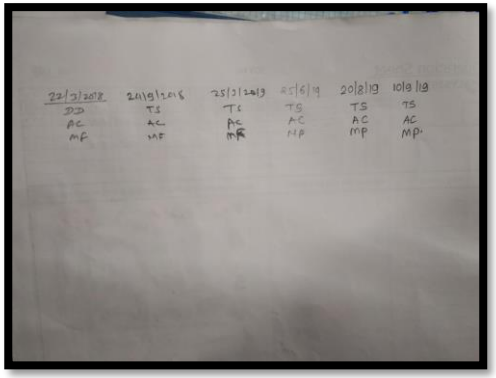
8.1 Updation of documents:

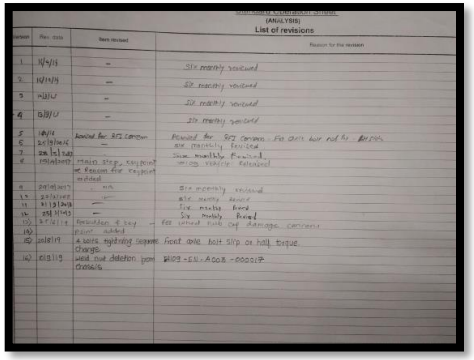
SOS Updated :- SCDC 4WD front axle drop on chassis

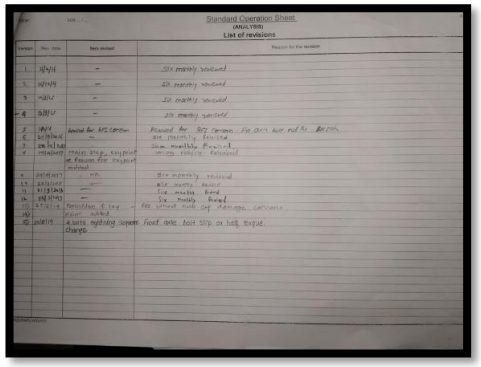












Work station SOS

8.1 Updation of documents:

- Control Plan – Front axle drop on chassis

Plant : Nashik		Model :Scorpio	Mahindra & Mahindra Ltd. Control Plan					
Part No/s.: - 2RD8SCHCB3T5EBR		Part Name/ Process: Chassis						
Latest Change Letter:-		Doc No.:	Prototype : ☐			Prelaunch: ☐		Production: ☐
Aggregate No.:		Aggregate Part Descrptn	Module Name : Scorpio Assy.			Reviewed by :Vikrant Ghule		Rev. No.: 171
PUI Shop :Scorpio/Xylo TCF		Next Aggregate No.:	Cell Name :Chassis			Approved by :Mandar Joshi		Rev. Date :20.09.19
Opn No.	Operation Description	Machine, Device, Jig, Tools Description	Characteristics			Sp. Char a. Clas s	Method	
			No.	Product	Process		Product/ Process Specification.	Evaluation Measurement Tech.
5690	Fitment Of FRT Axle & Lower Arm Assy using gauge lh & rh	Torque Controlled Nut Runner	1	Torque			63 ± 4 Nm	Torque Audit
			2		Lifting axle at ends and at center			Audit
		Hoist - 1 ton with lifting tackle	3		Condition of hoist & lifting tackle		As per PM checklist	Check list
			4	Axle ratio	Select axle as per buildsheet		1)Ratio 4.56(Diesel) 2)Ratio 4.88(Petrol) 3)Ratio 4.88 (LHD Petrol) 4) Ratio 4.55 SCDC"	Visual
			6		Calibration of GR tool		63 ± 4 Nm	Calibration as per plan
			7	Bolt head fitted with chassis bracket	Ensure bolt head matching with chassis bracket		.	Visual
		MS-Gague-14	8	Frt. Shock absorber to LCA gap	Use of gauge		As per SOS	Visual

8. Standardisation

8.2 Training:

Action Plan	Target Dt	Resp.	Status
Training given to stage associate for tightening of front axle bolts on chassis	08.09.18	Ajit Chaudhari	Closed

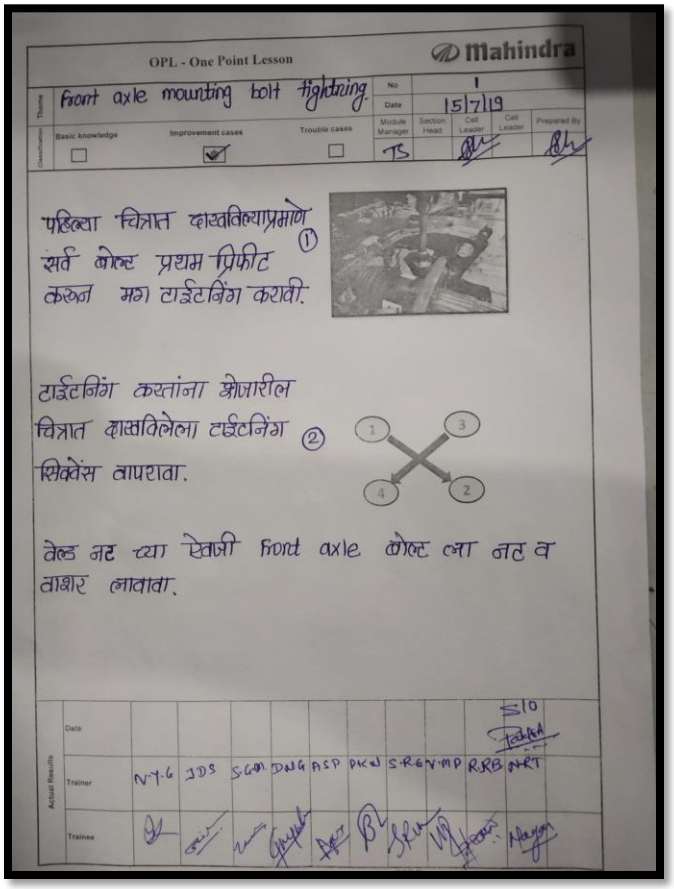
On Job training given to all stage associate for new tightening sequence

On Job training given to all stage associate for weld nut deletion

8. Standardisation

8.2 Training:

Action Plan	Target Dt	Resp.	Status
OPL given to all stage associates for new tightening sequence and weld nut deletion from chassis.	20.09.18	Ajit Chaudhari	Closed



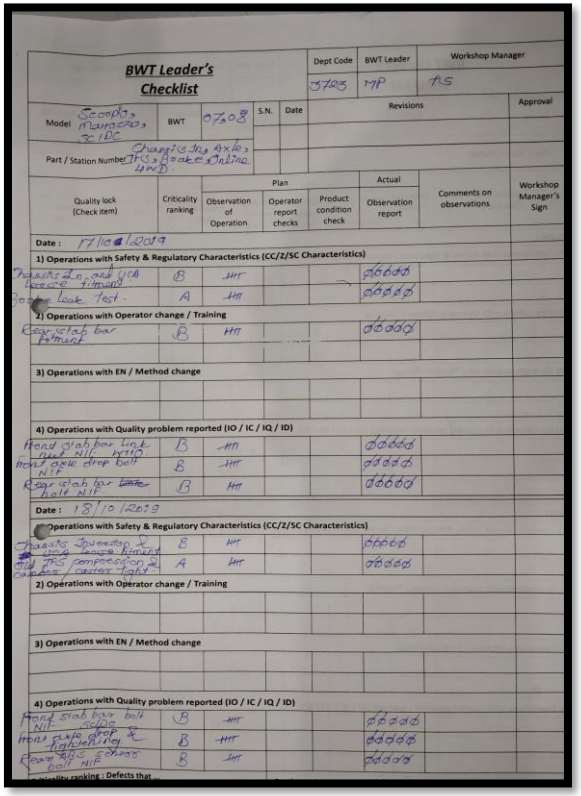
OPL given to all stage associate about front axle bolt tightening

8. Standardisation

8.3 Additional Audit / Check

Point taken in daily supervisory check sheet – till Oct'19

SEP 19



BWT Leader's Checklist

Dept Code	BWT Leader	Workshop Manager
3723	MP	TS

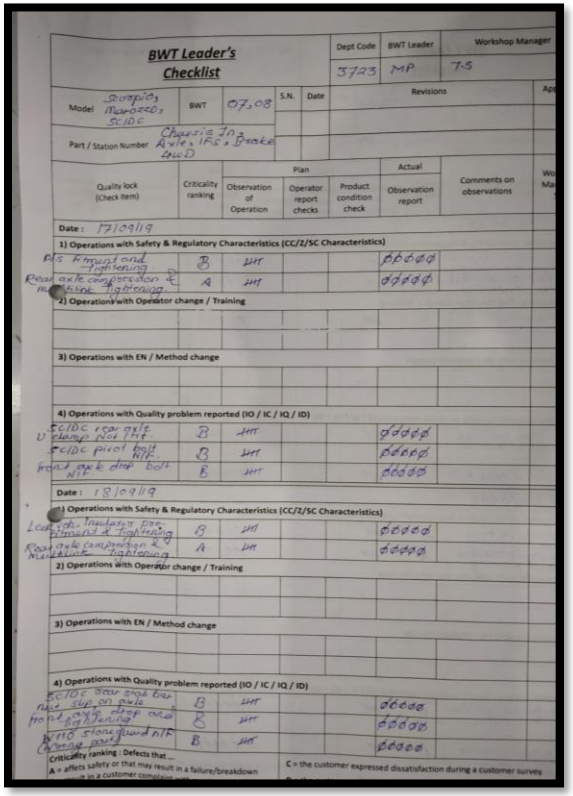
Model: Scorpio, Marazzo, SCIDC | BWT: 07.08 | S.N. | Date | Revisions | Approval

Part / Station Number: Chassis Jn, Axles, IFS, Brakes, Online, 4WD

Quality lock (Check item)	Criticality ranking	Plan: Observation of Operation	Plan: Operator report checks	Plan: Product condition check	Actual: Observation report	Comments on observations	Workshop Manager's Sign
Date : 17/10/2019							
1) Operations with Safety & Regulatory Characteristics (CC/Z/SC Characteristics)							
Chassis Jn and IFS bracket fitment	B	HH			ØØØØØ		
Brake leak test.	A	HH			ØØØØØ		
2) Operations with Operator change / Training							
Rear stab bar fitment	B	HH			ØØØØØ		
3) Operations with EN / Method change							
4) Operations with Quality problem reported (IO / IC / IQ / ID)							
Front stab bar link nut NIF, bush	B	HH			ØØØØØ		
Front axle drop bolt NIF	B	HH			ØØØØØ		
Rear stab bar ~~bolt~~ bolt NIF	B	HH			ØØØØØ		
Date : 18/10/2019							
1) Operations with Safety & Regulatory Characteristics (CC/Z/SC Characteristics)							
Chassis Jn weld & stab bar fitment	B	HH			ØØØØØ		
IFS compression & center tight	A	HH			ØØØØØ		
2) Operations with Operator change / Training							
3) Operations with EN / Method change							
4) Operations with Quality problem reported (IO / IC / IQ / ID)							
Front stab bar bolt NIF, scIDC	B	HH			ØØØØØ		
Front axle drop & tightening	B	HH			ØØØØØ		
Rear IFS stab bolt NIF	B	HH			ØØØØØ		

Criticality ranking : Defects that ...

OCT 19



BWT Leader's Checklist

Dept Code	BWT Leader	Workshop Manager
3723	MP	TS

Model: Scorpio, Marazzo, SCIDC | BWT: 07.08 | S.N. | Date | Revisions | Approval

Part / Station Number: Chassis Jn, Axles, IFS, Brakes, 4WD

Quality lock (Check item)	Criticality ranking	Plan: Observation of Operation	Plan: Operator report checks	Plan: Product condition check	Actual: Observation report	Comments on observations	Workshop Manager's Sign
Date : 17/09/19							
1) Operations with Safety & Regulatory Characteristics (CC/Z/SC Characteristics)							
IFS fitment and tightening	B	HH			ØØØØØ		
Rear axle compression & mounting tightening	A	HH			ØØØØØ		
2) Operations with Operator change / Training							
3) Operations with EN / Method change							
4) Operations with Quality problem reported (IO / IC / IQ / ID)							
SCIDC rear axle clamp not fit	B	HH			ØØØØØ		
SCIDC pivot bolt NIF.	B	HH			ØØØØØ		
Front axle drop bolt NIF	B	HH			ØØØØØ		
Date : 18/09/19							
1) Operations with Safety & Regulatory Characteristics (CC/Z/SC Characteristics)							
Lower arm, Insulator pin fitment & tightening	B	HH			ØØØØØ		
Rear axle compression & mounting tightening	A	HH			ØØØØØ		
2) Operations with Operator change / Training							
3) Operations with EN / Method change							
4) Operations with Quality problem reported (IO / IC / IQ / ID)							
SCIDC Rear stab bar nut, stop on axle	B	HH			ØØØØØ		
Front axle drop and tightening	B	HH			ØØØØØ		
RHD stone guard NIF, Orange paint	B	HH			ØØØØØ		

Criticality ranking : Defects that ...
A = affects safety or that may result in a failure/breakdown
C = the customer expressed dissatisfaction during a customer survey

8. Standardization

8.2 Additional Audit / Check

Process audit sheet modified for front axle drop on chassis.

PROCESS AUDIT SHEET BWT 7 & 8

BWT No.	Stage	Operation/Part	What to check	DATE							
7	AXLE	Traveller card	No filling of traveller card before stage.								
7		Build sheet	Vin no verificarion and barcode presence on buildhseet before pasting on peg.								
7		Power steering	bolt prefitment and sccanning for tightning								
7		Alluminium cradle	Ensure no gap between cradle and chassis mounting if found uses washer								
7		Hub fitment on RTB	Prefitment with hand tool,Scanning on computer,Ensure OK print after torquing,uses of locking pins on fixture before torquing,pasting of print on axle.								
7		Metal clip fitment on RTB	Orientation of clip and presence of clip on RTB								
7		Front axle drop on chassis	Tightning sequence as per SOS & no weld nut on chassis								
7		Bump stop fitment	Ensure prefitment of nut with special tool								
7		Metal cup placing on Chassis	Ensure metal cups between RTB and chassis at both side.								
7		Disc fitment on hub	Ensure metal lock on disc & No damage of disc & roating disc with no fouling.								
7		Caliper fitment	Ensure presence of brake shoes at both side , Bolt tightning with 50 Nm.								
7		Retorquing of hub bolts and caliper bolts LH & RH	Ensure retorquing of hub all 8 bolts with torque wrench at 100Nm & 4 Caliper bolts with torque wrench 50Nm.								
7		Axle bolt tightning	Ensure scanning before tightning at 200Nm								
7		Coil spring fitment	Ensure alignment of pad lug while coil spring mounting on italso ensure proper sitting of coil spring on pad and RTB. Match coil spring one end with chassis side member welding.								
7		Chain binder	Ensure positive locking of chain binders with rubber locks presence on it.								
7		Fuel tank & filter bracket fitment	Ensure tightning sequence & plain washer in all 4 boltsand orientation of fuel filter bracket.								
7		Rubber sleeve fitment on stab bar braket	Ensure fitment of rubber sleeve on bracket.								
7		Parking cable fitment	Ensure no damage of cable and rubber.Routine of cable , positive locking inside chassis.								
7	IFS	LCA loose fitment	Ensure Camber bolt as per model , & C clamp bolt prefitment 6-7 finger spins.								
7		Tightning of camber bolts, Cradle and C clamps	Ensure scanning before tightning . Ensure torque sequece - C clamp 100 & 100 then Cradle 300 & 300 and then last Camber bolt at 255 Nm. Ensure sound ande green signal after torque.								
7		Reinforcement plate fitment	Ensure bolt prefitment and no oil on weld nut.								
7		Brake tube clipping on chassis	Ensure positive locking of clips on chassis by pulling outward.								
7		Bundy tube locking inside clips	Ensure click sound for locking and pull after locking.								
7		Hand brake cable fitment on caliper	Ensure positive locking of cable inside caliper mounting.								
7		Spacer and Lock nut fitment	Ensure plain surface of wwasher towards lock nut.								

9. Summarise & Plan future actions

9.1 Conclusion:

Front axle bolt slip/half torque concern in SCDC model resolved by following actions.–

1. Front axle bolt mounting weld nut deletion from chassis.
2. Front axle all bolts tightening sequence changed.
3. Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random)

9.3 Horizontal Deployment:

1. NIL

9.3 Lessons Learnt:

THINGS GONE WRONG: At the time of W109 launch for pre engagement of axle weld nut was given but it was not benchmarked with existing W103 models.

THINGS GONE RIGHT: CFT worked in proper manner & resolved the concern & major rework eliminated and FRC increased.

Thank you……!